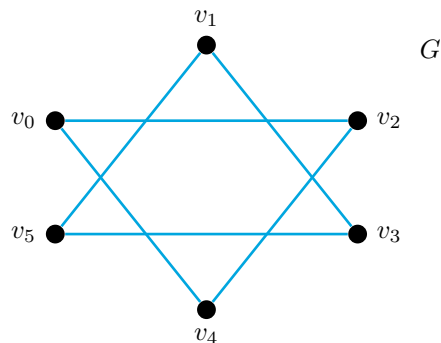


10.1.16

Determine which of the graphs in 12–17 have Euler circuits. If the graph does not have an Euler circuit, explain why not. If it does have an Euler circuit, describe one.



If a graph has an Euler circuit, then every vertex of the graph G has positive even degree.

$$\deg(v_0) = \begin{cases} v_0 - v_2 \\ v_0 - v_4 \end{cases} = 2$$

$$\deg(v_2) = \begin{cases} v_2 - v_0 \\ v_2 - v_4 \end{cases} = 2$$

$$\deg(v_4) = \begin{cases} v_4 - v_0 \\ v_4 - v_2 \end{cases} = 2$$

Similarly, $\deg(v_1), \deg(v_3), \deg(v_5)$ are equal to two. However, since the graph is **disconnected**: $\{v_2, v_3, v_4\}$ forms one triangle and $\{v_1, v_5, v_0\}$ forms another. The crossings in the drawing are not vertices. Therefore, the graph does not have an Euler circuit.

Solution: The graph does not have an Euler circuit.